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Stability and transferability of broadly trained phenology models in a changing climate

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ABSTRACT

A variety of phenology process-based models have been developed to simulate environmental influences on the timing of spring and autumn phenophases. Similar performances between different types of mechanistic models have raised questions about reliability of their predictions. To assess the biological relevance of phenology models, we used a seven-decade dataset of five species across 170 sites and 1700 m elevation in Switzerland. We evaluated nine leaf emergence and ten senescence models over time and space. We explored how optimal parameter values and influences vary, reflecting transitions in model aptitude and phenology responses to drivers. Leaf emergence models showed improved predictions at external sites over time, while emergence dates converged across Switzerland. In contrast, leaf senescence models often failed to outperform the null model predicting the mean date of training data and showed divergent performance trends. Trends in optimal parameters indicated species-specific responses to emergence drivers, with cold-climate suited species favouring earlier thresholds for warmth accumulation in spring, while the trends were opposite for warm-climate suited species, except for beech showing stable parameters likely due to strong photoperiod constraints. Warming increased the importance of chilling-related parameters for leaf emergence, while senescence parameter sensitivities remained stable. Spatial analyses revealed that complex models were less robust to training and validation at different elevations than simple models, and that phenological responses may vary non-linearly with elevation, likely due to local adaptations. Senescence models performed better with validation at high elevations, where climatic variables such as cooling temperatures play a large role, while predictions were more challenging at other elevations. These findings highlight the need for further refinement of process-based models to account for all driving influences on plant phenology, particularly for leaf senescence models. Our work demonstrates the potential for process-based modelling techniques to better understand phenology responses to climate change.

Data statement

All data pertaining to this article are available in the following repository: Mendeley Data, V1, doi:10.17632/j8ns6dsxck.1.

1. Introduction

1.1. Non-Linear plant phenology responses to climate change

Warming in the Anthropocene has altered leaf phenology, the timing of leaf life cycle events, by generally advancing spring leaf emergence and delaying autumn leaf senescence (IPCC et al., 2021). Under

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relatively stable historical climates, phenological strategies evolved to avoid seasonal disturbances such as frost while capitalizing upon growing season resources (Baumgarten et al., 2023; Pau et al., 2011). Therefore, the current pace of phenological shifts has far-reaching ecological and biogeochemical implications for forest ecosystems (Augspurger, 2013; Hufkens et al., 2012; Keenan et al., 2014; Vitasse and Rebetez, 2018; Wang et al., 2020a; Zohner et al., 2020). Over large regions, phenology patterns reflect latitudinal gradients, while in mountainous areas they follow elevational gradients (Meier et al., 2021; Moon et al., 2021; Wenden et al., 2020). Beyond a simple linear response, warming can both delay and advance leaf emergence. Warm winters with less exposure to chilling temperatures delay dormancy release, while warmer springs promote earlier dormancy release (Caffarra et al., 2011). Interactions between temperature and photoperiod also influence phenology. In cold springs lengthening photoperiods may compensate for insufficient warmth to stimulate leaf emergence, while in warm springs short photoperiods may constrain leaf emergence (Fu et al., 2019; Way and Montgomery, 2015). For instance, a study in Switzerland found that warming reduced phenological differences between elevations, presumably due to greater chilling exposure at higher elevations and increasing photoperiodic limitations at lower elevations (Vitasse et al., 2018a). Recent studies report a deceleration in the advancement of spring phenology and more consistent patterns in warmer springs (Chen et al., 2018; Denéchère et al., 2021; Fu et al., 2015; Piao et al., 2019). Nonetheless, climate variability has the potential to both advance and delay spring and autumn leaf phenology (Spafford et al., 2023a). To accurately predict future phenology, models must effectively simulate dormancy phases, interactions, and non-linear effects over long time periods. Currently, predictions remain uncertain as models with different process inclusions often perform similarly well under calibration conditions but diverge under future warmer conditions (Caffarra et al., 2011; Chuine et al., 2016; Häkkinen et al., 1998; Kramer, 1994; Meier and Bigler, 2023; Zhao et al., 2021).

1.2. Phenology process models

Models simulating the causal relationship between seasonal fluctuations in environmental drivers and physiological phenology mechanisms are known as process-based models of phenology (Chuine, 2000; Hänninen, 2016). Process-based models have been in use for centuries, such as the growing degree day model developed by de Reaumur (1735) for predicting the phenology of grape plants. Since the 18th century, process-based models have incorporated a detailed representation of phenology processes and responses gleaned from experiments, including multiple interactive environmental drivers and internal developmental processes. For example, the Dormphot model includes the phases ofendodormancy, ecodormancy, and dormancy induction (Caffarra et al., 2011; Coville, 1920; Piao et al., 2019). Incorporating numerical representations of these processes and interactions into models has allowed for the simulation of non-linear effects as well as predictions of changes in phenology over long periods of time (Lange et al., 2016; Spafford et al., 2023b; Vitasse et al., 2011). Anticipating long-term phenological changes through accurate process-based models can provide valuable insights for ecological management, land-atmosphere modelling, and carbon sequestration projections (Keenan et al., 2014; Kowalski et al., 2020; Norton et al., 2023; Richardson et al., 2013; Tang et al., 2016).

1.3. Model complexity

Increasingly complex models, with more parameters and drivers, offer both advantages and disadvantages over more simple process-based models. For example, the Dormphot model offers a realistic representation of phenological drivers and their interactions based on experiments manipulatingphotoperiod, chilling, and forcing (Caffarra et al., 2011), potentially improving long-term predictions under changing conditions. Model intercomparison studies have found that complex models with two or three dormancy phases outperformed simpler models under anomalous conditions, and with improved transferability over time at a given site (Basler, 2016; Liu et al., 2019; Schädel et al., 2023; Spafford et al., 2023b; Teets et al., 2023). In contrast, simpler models, with just one or two drivers, often transfer better to external sites than complex models and generalize better across large datasets (Basler, 2016; Gao et al., 2024; Melaas et al., 2016; Schädel et al., 2023). Other important factors to consider in the compromise between complex and simple models include efficiency and overfitting. The calibration of complex models with a greater number of parameters requires greater computational resources relative to simpler models. Complex models also necessitate longer meteorological records prior to leaf emergence for capturing the phases of dormancy induction and endodormancy. In addition, models with a greater number of parameters are more susceptible to overfitting (Burnham and Anderson, 2004). Consequently, model intercomparison studies would benefit from the assessment of a range of models, including simple, intermediate, and complex models.

1.4. Limitations in model predictions

Notably, despite markedly differing mechanisms, one phase and three-phase models perform similarly(Basler, 2016; Hufkens et al., 2018; Spafford et al., 2023b; Teets et al., 2023). In combination with divergent outcomes between similar model types, this raises questions about the reliability of predictions derived from these models (Zhao et al., 2021). A key challenge is that many different parameter combinations within models can yield similar results without biological relevance. For example, Wang et al. (2020b) found that several leaf emergence process-based models include an unrealistic positive relationship between chilling accumulation and heat requirement for ecodormancy release, leading to overestimated leaf emergence advances. Further, a recent study by Gao et al. (2024) demonstrates that common process-based models may overemphasize the role of chilling, leading to underestimations of emergence advancement in response to warming. Sensitivity analyses highlight potential shifts in driver influences with warming, such as an increasing influence of chilling and interactions with photoperiod (Gauzere et al., 2019; Lange et al., 2016). Additionally, the greater leaf emergence advancement for species growing at high elevations with warming due to ample chilling exposure at high elevations may lead to an earlier onset and accumulation of chilling and forcing influences for some species, altering largescale patterns in dormancy response processes. Consequently, the relative importance of process-based model components, and their overall utility, may evolve with changes in conditions over time. Phenological sensitivity to warming is currently the subject of an important debate for predictions and model development, as it is unclear whether it will decline or remain constant (Fu et al., 2015; He et al., 2023; Wolkovich et al., 2021). Along with an accurate depiction of phenological sensitivity to warming, the improvement of autumn leaf phenology models is a key priority (Gallinat et al., 2015), as these models often exhibit twice the error of spring models for the same species and conditions (Liu et al., 2020; Meier and Bigler, 2023; Spafford et al., 2023b; Zani et al., 2020). Many autumn phenology models are numerically like the inverse of spring phenology models, despite observational and experimental studies revealing that other cues in additional to cumulative environmental cues play an important role (Bigler and Vitasse, 2021a; Gan and Amasino, 1997; Vitasse et al., 2021). Recently, the integration of previous budburst timing during spring has led to improved autumn predictions in several studies (Fu et al., 2014; Keenan and Richardson, 2015; Zohner et al., 2023).

1.5. Knowledge gaps and research questions

Mechanistic or process-based models can be used as a tool to better understand plant phenology responses to environmental changes

(Chuine et al., 2013; Chuine and Régnière, 2017; Meier and Bigler, 2023). Despite this, few studies have assessed the stability and biological relevance in modelled plant phenology responses to phenology-driving processes over time (Fu et al., 2015). Along with long-term data, plant phenology modelling techniques could be leveraged to glean trends in plant phenology responses to phenology-driving processes over time, assess the biological relevance of mechanistic representations within process-based models, and to better understand plant phenology processes. If plant phenology departs from expected trends over time in response to warming, such as recent observations of reduced variability in phenology in response to warming (Denéchère et al., 2021), this would have important implications for projections in the context of climate change. Further, this departure may implicate declines in model aptitude over time, and current models may be less applicable to predicting future leaf phenology patterns in the context of climate change. Warming may alter the relative influence of model drivers (Gauzere et al., 2019), and shift the optimal values of model parameters over time. While most influential parameters would be expected to have relatively stable optimal values over time, though there is limited evidence on which to base this expectation. Here, we aim to address these knowledge gaps with a unique long-term tree phenology observational dataset collected across a broad range of climatic conditions for more than seven decades. We assess whether plant responses to phenology-driving processes have changed over time by evaluating the performance of process-based models 1) over time and 2) space, as well as 3) how optimal parameter values vary over time. Further, we examine potential changes in the relative importance of meteorological drivers by 4) quantifying the direct and interactive influence of model parameters on model outcomes.

2. Materials and methods

2.1. Phenology data

To calibrate our leaf phenology models, we used long-term data collected across Switzerland by the volunteer-based Swiss Phenology Network (SPN), a monitoring program managed by MeteoSwiss since 1951 (Fig. 1). Long-term leaf emergence records covering >50 years were available from 52 or more sites for each of five species. Two species are suited to cold climates: Norway spruce (*Picea abies*), and European larch (*Larix decidua*), while the other three are suited to warmer climates: European beech (*Fagus sylvatica*), horse chestnut (*Aesculus hippocastanum*), and hazel (*Corylus avellana*). Long-term leaf senescence (colouration) records were only available for European beech and horse chestnut. Observational sites were located between 200 and 1930 m above sea level and across an annual average temperature gradient from

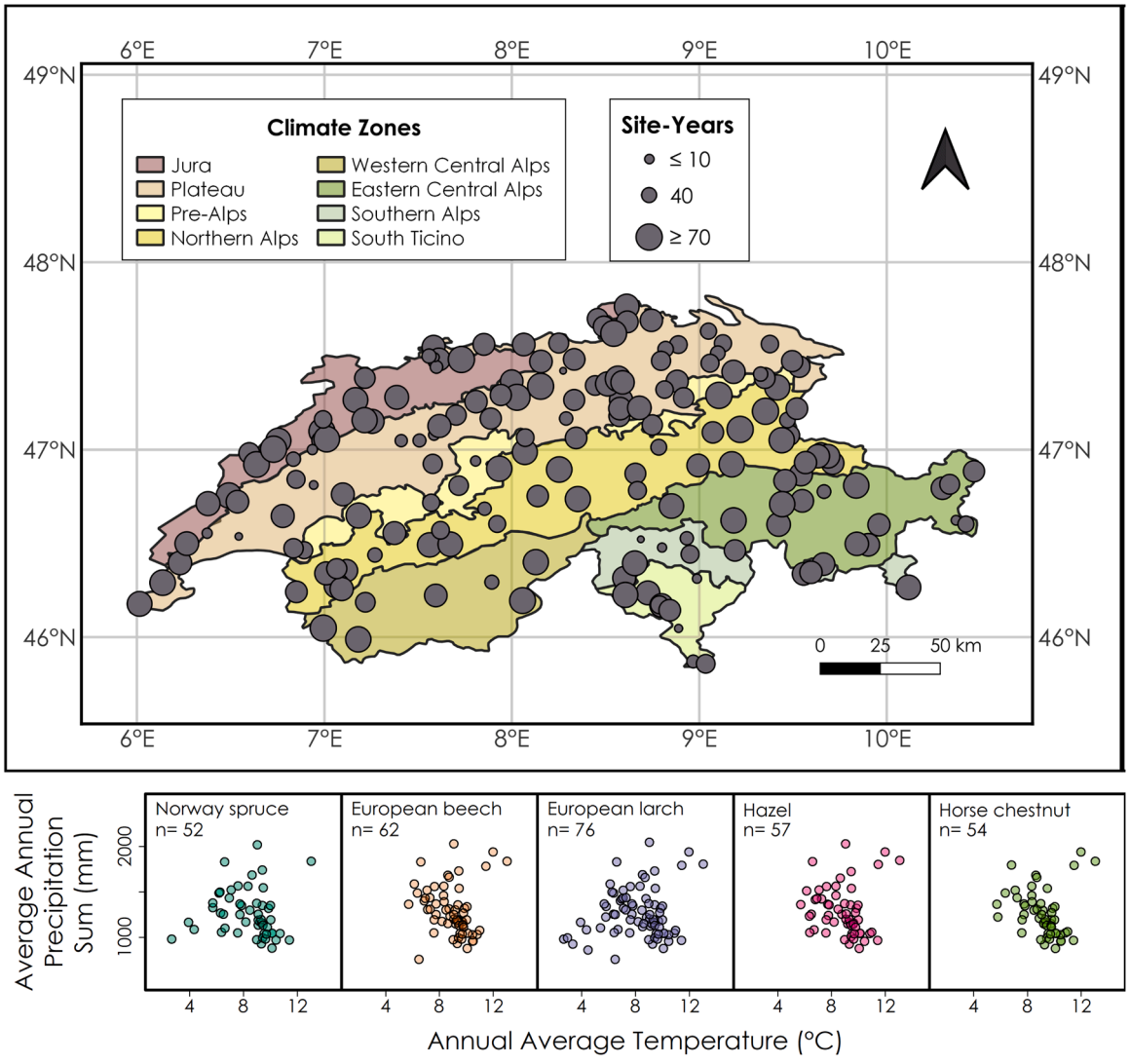


Fig. 1. Location and duration of MeteoSwiss observation sites across climate zones of Switzerland as defined by MeteoSwiss (top). The average annual precipitation and temperature from 1991–2020 at each long-term operational (>50 y) site for each species, with the number of sites for each species indicated (bottom).

1.5 to 14.1 °C. There was a narrow photoperiod gradient across sites from 13.3 to 13.4 h on April 15th, and 10.7 to 10.8 h on October 15th.

At each site, trees were observed one to three times a week by one or more observers and their phenophase was recorded (Bigler and Bugmann, 2018; Güsewell et al., 2018; Vitasse et al., 2018a). Leaf unfolding is defined as the date when 50 % of the leaves or needles are unfolded or have emerged. Leaf colouration is defined as the date when 50 % of the leaves have begun to exhibit autumn colours or 50 % of the leaves have already fallen off.

2.1.1. Quality control

Several steps were taken to account for potential observational errors, observer disagreements, and outliers due to local anomalies (e.g. disease, insect pests), detailed in Auchmann et al. (2018). During the project PhenoClass (2016–2018), an automatic flagging protocol was developed to remove outliers. Following this automatic flagging, two experts reviewed flagged values to preserve valid observations resulting from extreme weather. To ensure equal temporal representation of phenology for our analyses, we further removed sites with <50 years of observations, resulting in 17,634 total leaf emergence observations across the five long-term monitored species and 5160 total leaf colouration observations.

2.2. Climate data

Daily average, minimum, and maximum temperature, along with daily precipitation sums for all sites were obtained from spatially interpolated weather station data made available from 1930 to 2022 by MeteoSwiss. The Daymet interpolation algorithm from Thornton et al. (1997) was applied to this station data along with a 5-m digital elevation model to account for the steep elevational gradients across Switzerland. The resulting 100-m resolution gridded data product was provided by the Landscape Dynamics Lab at the Swiss Federal Institute for Forest, Snow, and Landscape Research. Daily photoperiod was computed using the daylength() function in the package phenor (Hufkens et al., 2018).

2.3. Phenology models

2.3.1. Leaf emergence models

We used eight leaf emergence process-based models with varying levels of complexity and process representation, along with a single statistical model from the phenor package (Hufkens et al., 2018; Table 1). Each process-based model adheres to the same general form: following a start date for accumulation, they simulate an accumulation of driver influence until a critical threshold is reached and leaf emergence occurs. Four process-based models simulate ecodormancy release only, triggered either by forcing spring temperatures (Thermal Time, TT) or by forcing and lengthening photoperiod (Photo-Thermal Time, PTT; Photo-Thermal Time with Sigmoidal Temperature Response, PTTs; M1) as drivers. While the TT model does not include an explicit photoperiod driver, the start date parameter may imply an inherent photoperiod constraint (Basler, 2016). Three process-based models simulate both endodormancy and ecodormancy, with chilling winter temperatures and forcing spring temperatures (Alternating, AT; Parallel, PA; Sequential, SQ). In addition to these, one more comprehensive process-based model simulates dormancy induction along with endodormancy and ecodormancy (Dormphot, DP). Note dormancy induction is distinct from paradormancy, which occurs after bud set in late summer or early autumn and is hormonally regulated by distal organs rather than environmental variables (Delpierre et al., 2016). We also included one statistical model (LIN) based on average spring temperature from March 1st to May 31st (day of year 60 to 151) and a null model that assumes the mean emergence date of all training samples, for comparison with process-based models (Basler, 2016). Equations of all models are available in Appendix A1 as well as through the references cited in Table 1.

2.3.2. Leaf senescence models

We incorporated ten leaf senescence process-based models to simulate autumn leaf colouration, spanning a range in complexity (Table 2). Most models simulate an accumulation of cooling temperatures and shortening daylength influences until a critical threshold is reached and leaf senescence (colouration) occurs. One model, the White Model (WM), uses an instant triggering mechanism instead of accumulation, based on either photoperiod and cooling or cooling alone. Five models only simulate dormancy induction, though with varying forms (WM; Delpierre 1, DM1; Delpierre 2, DM2; Delpierre 1-Zani 2020, DM1Za20; Jeong Model, JM; Dormphot Dormancy Induction, DPDI). Two models incorporate preceding spring leaf emergence, simulated through the TT model, assuming an earlier leaf emergence corresponds with an earlier leaf senescence (Delpierre 1 with Preceding Spring Leaf Emergence, DM1s; Dormphot Dormancy Induction with Preceding Spring Leaf Emergence, DPDIs). The Photosynthesis-Influenced Autumn with Growing Season Index (PIAG) recently developed by Zani et al. (2020)

Table 1
Model names along with their simulated processes and drivers for leaf emergence included in this study arranged from the simplest to the most complex based on the number of free parameters.

Model	Process(es) included	Driver(s) included	Number of free parameters	Reference
Linear (LIN)	None (statistical)	Forcing	2	Basler, 2016; Donnelly et al., 2006; Hufkens et al., 2018
Thermal Time (TT)	Ecodormancy Release	Forcing	3	Basler, 2016; de Reaumur, 1735; Hufkens et al., 2018; Wang, 1960
Photo-Thermal Time (PTT)	Ecodormancy Release	Forcing & Photoperiod	3	Basler, 2016; Črepinšek et al., 2006; Hufkens et al., 2018; Masle et al., 1989
Photo-Thermal Time with Sigmoidal Temperature Response (PTTs)	Ecodormancy Release	Forcing & Photoperiod	4	Basler, 2016; Črepinšek et al., 2006; Hänninen, 1990; Hufkens et al., 2018; Kramer, 1994; Masle et al., 1989
M1	Ecodormancy Release	Forcing & Photoperiod	4	Basler, 2016; Blümel and Chmielewski, 2012; Hufkens et al., 2018
Alternating (AT)	Endodormancy & Ecodormancy Release	Chilling & Forcing	5	Basler, 2016; Cannell and Smith, 1983; Hufkens et al., 2018; Murray et al., 1989
Sequential (SQ)	Endodormancy & Ecodormancy Release	Chilling & Forcing	8	Basler, 2016; Hänninen, 1990; Hufkens et al., 2018; Kramer, 1994; Landsberg, 1974
Parallel (PA)	Endodormancy & Ecodormancy Release	Chilling & Forcing	9	Basler, 2016; Hänninen, 1990; Hufkens et al., 2018; Kramer, 1994
Dormphot (DP)	Dormancy Induction, Endodormancy & Ecodormancy Release	Cooling, Chilling, Forcing, & Photoperiod	11	Basler, 2016; Caffarra et al., 2011; Hufkens et al., 2018

Table 2
Model names along with their simulated processes and drivers for leaf senescence included in this study arranged from the simplest to the most complex based on the number of free parameters.

Model	Process(es) Included	Driver(s) Included	Number of Free Parameters	Reference
White (WM)	Dormancy Induction	Cooling & Photoperiod	3	Liu et al., 2020; White et al., 1997
Delpierre 1-Zani 2020 (DM1Za20)	Dormancy Induction	Cooling & Photoperiod	3	Delpierre et al., 2009; Liu et al., 2020; Meier and Bigler, 2023; Zani et al., 2020
Jeong (JM)	Dormancy Induction	Cooling & Photoperiod	3	Jeong and Medvigy, 2014; Liu et al., 2020
Dormphot Dormancy Induction (DPDI)	Dormancy Induction	Cooling & Photoperiod	4	Caffarra et al., 2011; Hufkens et al., 2018; Liu et al., 2020
Delpierre 1 (DM1)	Dormancy Induction	Cooling & Photoperiod	5	Delpierre et al., 2009; Liu et al., 2020; Meier and Bigler, 2023; Zani et al., 2020
Delpierre 2 (DM2)	Dormancy Induction	Cooling & Photoperiod	5	Delpierre et al., 2009; Liu et al., 2020; Meier and Bigler, 2023; Zani et al., 2020
Dormphot Dormancy Induction with Preceding Spring Leaf Emergence (DPDIs)	Preceding Ecodormancy Release & Dormancy Induction	Forcing, Cooling, & Photoperiod	5	Caffarra et al., 2011; Liu et al., 2020
Photosynthesis-Influenced Autumn with Growing Season Index (PIAG)	Preceding Ecodormancy Release, Growing Season Productivity & Dormancy Induction	Forcing, Cooling, Precipitation & Photoperiod	5	Meier and Bigler, 2023; Zani et al., 2020
Delpierre 1 with Preceding Spring Leaf Emergence (DM1s)	Preceding Ecodormancy Release & Dormancy Induction	Forcing, Cooling, & Photoperiod	6	Delpierre et al., 2009; Liu et al., 2020
Delpierre Sigmoidal with Precipitation Influence (DMP)	Preceding Ecodormancy Release & Dormancy Induction	Forcing, Cooling, Precipitation & Photoperiod	6	Delpierre et al., 2009; Meier and Bigler, 2023; Zani et al., 2020

includes the effect of estimated productivity, assuming a more productive growing season leads to earlier leaf senescence. For the PIAG model, the estimated productivity index is computed separately following the timing of leaf emergence estimated with the TT model until daylength reaches 11 h in the autumn, at which point photosynthesis no longer occurs (Meier and Bigler, 2023; Zani et al., 2020). The Delpierre Sigmoidal with Precipitation Influence (DMP) model simulates the influence of low precipitation days (< 2 mm) as well as average growing season temperature on senescence timing, from the timing of leaf emergence estimated with the TT model until daylength reaches 11 h in the autumn (Delpierre et al., 2009; Meier and Bigler, 2023; Zani et al., 2020). Note for both the PIAG and DMP models, the influence of cooling temperatures and shortening photoperiod accumulates starting on July 1st, while other drivers (productivity and precipitation, etc.) influence the magnitude of critical thresholds and are accumulated separately following the timing of estimated leaf emergence. Unlike the leaf emergence models, all these models have a fixed start date of July 1st for the accumulation of cooling and shortening photoperiod effects following the summer solstice. Equations of all models are available in Appendix A1 as well as through the references cited in Table 2.

2.3.3. Model calibration

The optimal parameters for each model calibration (training) were selected with general simulated annealing (GenSA) via the pr_fit_parameters() function in the R package phenor, which invokes the GenSA optimization function within the GenSA package in R (Xiang et al., 2013). We chose this calibration approach as a study comparing several approaches for the calibration of phenology process-based models found this approach to yield the best-performing models (Meier and Bigler, 2023). General simulated annealing searches for optimal model parameter values while converging towards a global minimum in model prediction error in an analogous fashion to the process of metal cooling (Chuine et al., 1998). Unlike other calibration algorithms, such as the Particle Swarm Optimization, or Covariance Matrix Adaptation with Evolutionary Strategies, GenSA permits the acceptance of suboptimal parameter values during optimization, reducing the risk of confinement in local minima (Meier and Bigler, 2023). We configured calibrations to seek the lowest global root mean squared error (RMSE) between modelled and observed phenology with a starting temperature of 10,000 and a maximum of 40,000 model calls, following Hufkens et al. (2018). This high starting temperature allows for thorough exploration of the parameter space while avoiding premature confinement in local minima.

Parameter ranges were established based on previous studies as well as the growing conditions of our study, and are provided in Appendix A1 (Basler, 2016; Hufkens et al., 2018; Liu et al., 2020; Zani et al., 2020). For leaf emergence models, the default parameter ranges from the phenor package were used, with two exceptions: the minimum base temperature for forcing accumulation was set to 0 °C, and the earliest start date for forcing accumulation was set to January 1st to ensure biologically relevant forcing parameters (Chuine et al., 2013; Chuine and Régnière, 2017; Delpierre et al., 2016). For leaf senescence models, the parameter ranges from Zani et al. (2020) and Meier and Bigler (2023) were used, with narrower ranges for daylength (8–16 h), and cooling base temperature thresholds (10–30 °C) to reflect possible ranges in our meteorology data.

2.3.4. Stratification of training and validation datasets

To ensure a uniform distribution of climate conditions in our training and validation datasets where intended, we established stratified datasets following the recommendations of Meier and Bigler for all our temporal model performance evaluations (2023), though based on long-term mean seasonal temperatures at each site rather than annual average temperatures. For leaf emergence, mean seasonal temperatures were calculated from September 21st of the previous year until May 1st of the year of leaf emergence, and then averaged for all available years of

observations for each site from 1951 to 2022 to produce a single long-term mean seasonal temperature value for each site. An equal number of sites for training and validation were then chosen from each of three groups of data (bins) based on long-term mean seasonal temperatures at each site: ≤3 °C, >3 and ≤5 °C, and >5° For leaf senescence, mean seasonal temperatures were calculated from May 1st to October 1st of the year of leaf colouring, and then averaged for all years of observations for each site to produce a single long-term mean seasonal temperature value for each site. An equal number of sites for training and validation were then chosen from each of three bins based on long-term mean seasonal temperatures at each site: ≤14 °C, >14 and ≤16 °C, and >16°

We selected these bins to ensure the greatest possible number of sites in each bin for all species. This resulted in at least four sites from each bin being available for training and at least four for validation. For the temporal trend evaluations presented below, we were interested in 1) how well models performed over time in terms of transferability to sites outside the training dataset, as well as 2) transferability to sites outside the training dataset and years beyond the range of the training dataset. For both temporal evaluations, sites included in the training data were not included in the validation data. We refer to these sites as external sites throughout, following the convention of other phenology process-based modelling studies (Basler, 2016; Chuine, 2000; Chuine et al., 1999, 1998).

2.3.5. Model evaluations

2.3.5.1. Evaluation metrics. To quantify model performances, we calculated the root mean square error (RMSE) between observed and predicted values. To test for significant differences in model performances within species we used the non-parametric Kruskal-Wallis test and compared model performances using the Dunn's post-hoc test. We also computed the Akaike's Information Criterion corrected for small samples (AICc) using the phenor package in R to account for model complexity for temporal evaluations as follows:

$$AICc = 2 * k + n * \log\left(\frac{RSS}{n}\right) + \frac{2 * k * (k+1)}{n-k-1} \quad (1)$$

where k is equal to the number of model parameters, n is equal to the number of observations, and RSS is the residual sum of squares between observations and predictions. We also calculated RMSE and AICc for the null model which assumed a fixed mean date based on training samples for comparison with the performances of process-based models. We considered a model to perform well if it outperformed the null model by at least 3 days.

2.3.5.2. Temporal trends in model performance. To assess the spatial and temporal transferability of models over time, we conducted several performance evaluations with stratified training and external validation site observations as follows. These evaluations along with all other evaluations are summarized in Fig. 2.

1. Spatial transferability over time: Trained models with 30-year moving windows from the training sites and validated with the same 30-year moving windows of observations at the external validation sites, and then quantified models' performances based on RMSE and AICc for validation sites. For this evaluation, we were interested in examining whether the spatial transferability of models is changing over time, which is why we used the same 30-year time periods for training and validation though with external sites not included in the training dataset. A period of 30 years was chosen to account for interannual variability and to be consistent with the length of period used for the calculation of climate metrics (Arguez and Vose, 2011). In addition, a 30-year training period helps to ensure that the outcome of this evaluation reflects model transferability rather than the influence of a limited training duration. In other studies, training periods and site inclusion thresholds for model training are typically <30 years (Caffarra et al., 2011; Meier and Bigler, 2023).
2. Spatial transferability and timeseries applicability over time: Trained models with 30-year moving windows from the training sites and validated with all available years of observations at the external validation sites, and then quantified models' performances based on RMSE and AICc for validation sites. For this evaluation, we were interested in the models' ability to predict phenology at external sites over the entire seven-decade period and how this varied depending upon the training window.

2.3.5.3. Calibration trends and parameter sensitivity. We also examined trends in calibrated parameters to determine how plant responses to phenology-driving processes may have changed over time. In addition, we quantified parameter sensitivity to determine the relative influence of parameters on model outcomes. These evaluations along with all other evaluations are summarized in Fig. 2.

1. Calibration trends: Held all except one of the parameters constant at the optimal values determined from a long-term calibration using all site-years from the stratified training pool of sites with four sites from each bin. We then iteratively determined the optimal value of freely varying parameters with 30-year moving windows and assessed any potential trends in parameters based on Spearman rank correlations with time.
2. Parameter sensitivity: Computed the first order and total-effects sensitivity Sobol Jansen indices for each model parameter using the soboljansen() function from the sensitivity package in R (Iooss et al., 2023) following Gauzere et al. (2019) with 10^5 iterations. First order sensitivity indicates the direct individual impact of each parameter on model outputs. Total-effects sensitivity indicates the impact of a parameter's interactions with other parameters (indirect effects), along with its direct individual impact. To examine how sensitivity varies under different seasonal conditions, we computed these indices for each site-year of observations from a stratified training pool of sites with three sites from each bin. We established our bins with three sites rather than four sites for this exercise to limit computational intensity.

2.3.5.4. Spatial patterns in model performance. To assess the spatial transferability of models irrespective of time, we trained and validated models with observations from a range of elevations. For these evaluations, we were interested in how phenology responses may differ between elevations, rather than general responses across the climate of Switzerland. These evaluations along with all other evaluations are summarized in Fig. 2.

1. Influence of training elevation - spatial transferability with stratified validation: Trained all models with an equal number of randomly sampled observations from each elevational band, low (< 600 m), medium (600 – 1000 m), and high (> 1000 m), and then validated with all records from stratified validation sites (4 from each bin) to test model predictions across a wide range of conditions, and then quantified models' performances based on RMSE for stratified validation sites.
2. Influence of validation elevation - spatial transferability with stratified training: As the inverse of 1), we trained all models with all records from stratified training sites (4 from each bin) and then validated with an equal number of randomly sampled observations from each elevational band, low (< 600 m), medium (600 – 1000 m), and high (> 1000 m) to test broadly trained model specificities to particular elevations, and then quantified models' performances based on RMSE for each validation elevation.

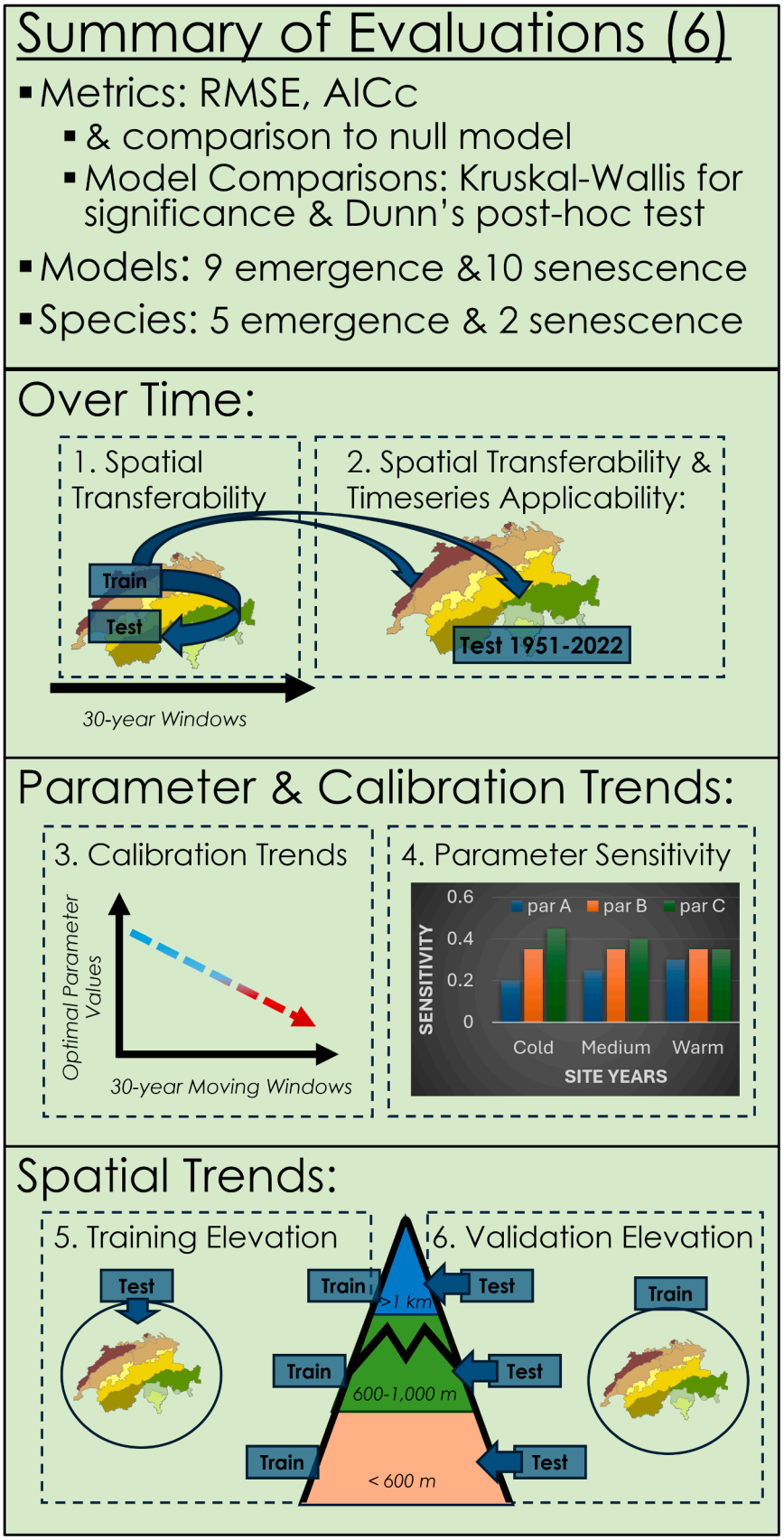


Fig. 2. Conceptual summary of all evaluations in this study. The number of training and validation site-years for each evaluation are shown in Table A1.5.

3. Results

3.1. Changes in phenology and growing conditions over time

Since 1951, leaf emergence has advanced at all elevations for all five long-term monitored species, with the greatest advancement being at higher elevations (Table A2.1). In contrast, leaf colouration followed a mixed pattern across species and elevations. An analysis of covariance (ANCOVA) revealed a highly significant interaction between species-level leaf emergence and seasonal temperature ($F_{4,17,624} = 66.1$, $p < 0.0001$), indicating that the phenological response to warming differed among species. Species with the strongest leaf emergence advancement in response to warming included (±SE): European larch (5.02 ± 0.06 days per °C), hazel (5.03 ± 0.09 days per °C), and Norway spruce (4.91 ± 0.08 days per °C) which we confirmed with estimated marginal trends using the emtrends() function in R (Lenth, 2024). The advancement in response to warming was not significantly different between these species ($p > 0.05$). In contrast, the leaf emergence of European beech was the least sensitive to warming (~3.4 ± 0.09 days per °C), and significantly differed from all other species ($p < 0.0001$; Fig. 3). Additionally, horse chestnut showed a reduced rate of advancement (4.6 ± 0.09 days per °C) compared to European larch ($p < 0.01$) and hazel ($p < 0.01$), but was not significantly different from Norway spruce ($p > 0.05$). Residuals met the assumption of homogeneity of variances, as assessed by a residuals vs. fitted values plot.

3.2. Temporal trends in model performance

3.2.1. Predicting and training leaf emergence models with the same moving windows at external validation sites

All leaf emergence models performed well when using only 30 years of training in predicting leaf emergence dates at the external validation sites over the same 30-year period, outperforming the null model and demonstrating spatial transferability (Fig. 4 and Figure A3.1). This evaluation revealed distinct performance patterns over time between species based on RMSE. For European beech and horse chestnut, process-based model spatial transferability improved over time, with RMSE values declining by about 5 and 9 days, respectively, when training and validating with more recent decades. Norway spruce saw an initial decline in RMSE values followed by an increase, while RMSE values of hazel and larch showed a slight decrease over time. RMSE trends correlated strongly with the standard deviation of observations at validation sites for all species ($r = 0.8$ - 0.99; $p < 0.001$) except Norway spruce ($r = 0.54$; $p < 0.001$), which is demonstrated in Figure A3.2. All models exhibited a non-linear pattern in AICc over time (Figure A3.1), with an initial increase followed by a decline. Variables underlying the AICc, the number of validation window observations over time, and the residual sum of squares, both followed a unique non-linear trend over time.

For all temporal trend evaluations, to determine if there were significant differences in model performances, we used a Kruskal-Wallis test and Dunn's post-hoc test to compare models, as our RMSE values were not normally distributed. Kruskal-Wallis tests indicated differences between models were significant for all species ($p < 0.001$). Dunn's post-hoc test comparisons confirmed that the null model had significantly higher RMSE than all other models for all species. Some intermediate complexity models incorporating either photoperiod (e.g. M1, PTT, PTTs) or chilling effects, (e.g., PA, AT) often significantly outperformed the DP, LIN, and SQ models for all species except horse chestnut based on Dunn's post-hoc test comparisons. In both AICc and RMSE evaluations, simple and intermediate leaf emergence models often outperformed more complex models, though no single model was consistently superior across species, which is in line with previous model intercomparison studies (Basler, 2016; Melaas et al., 2016; Schädel et al., 2023).

3.2.2. Predicting all seven decades of leaf emergence at external validation sites with moving training windows

When predicting all seven decades of leaf emergence at external validation sites, all process-based models outperformed the null model across different 30-year training windows based on RMSE and AICc (Figure A3.3 and Figure A3.4), demonstrating transferability over both time and space. Model performances were stable regardless of training period over seven decades, except for horse chestnut, which showed

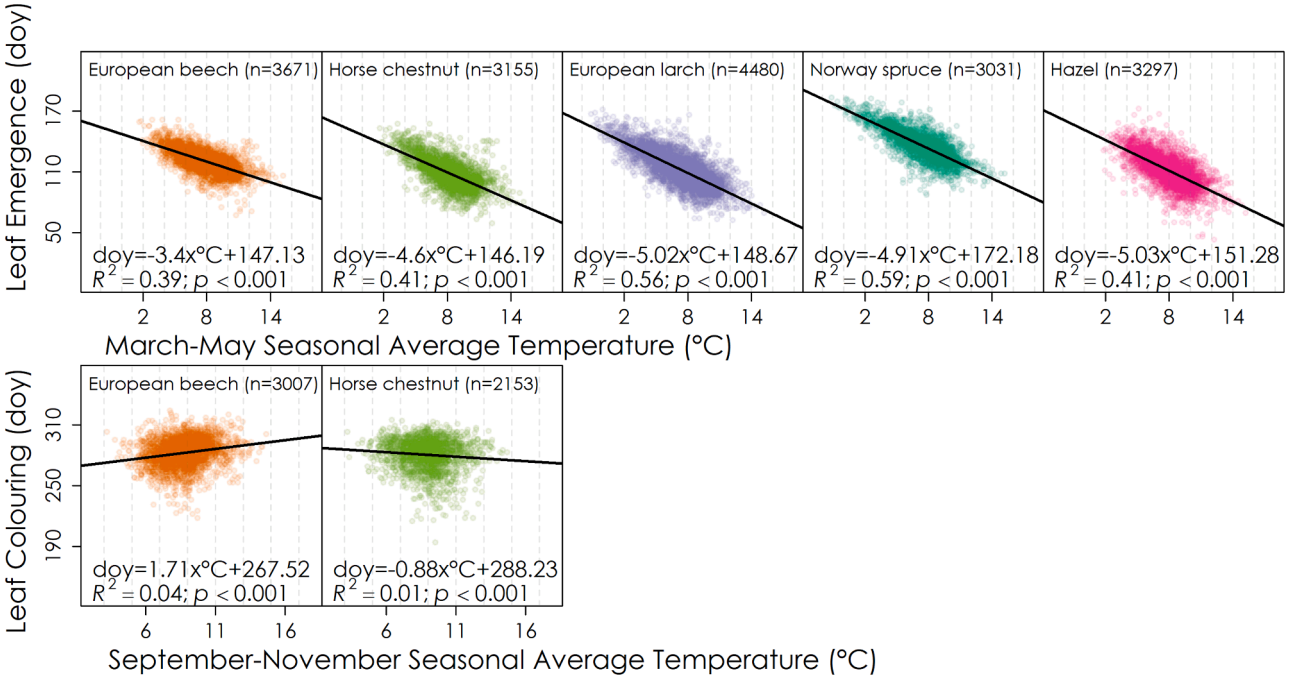


Fig. 3. Leaf emergence advances in response to warmer spring (MAM) temperatures for the five long-term monitored species in the Swiss Phenology Network. Conversely, leaf colouration exhibits no clear response to autumn (SON) temperature for the two species with long-term leaf colouration observations available.

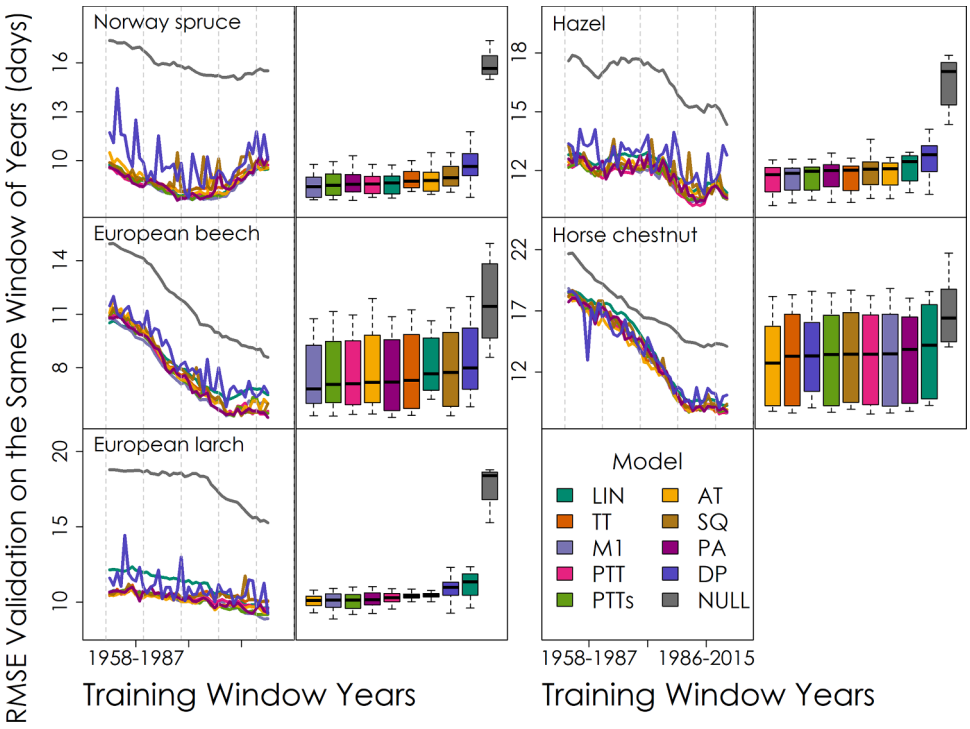


Fig. 4. Line plots showing performance of leaf emergence models with 30-year moving training windows in predicting leaf emergence dates at external validation sites for the same 30-year period, based on root mean squared error. Refer to Table 1 for model abbreviations. Boxplots to the right of each line plot show the overall performance over time for each species-model combination.

slight improvement with later training windows. The null model's performance varied between species and with training windows. For European larch and hazel the null model showed improvement with later training windows while remaining stable or slightly worsening over time for other species. Kruskal-Wallis tests indicated differences between models were significant for all species for this evaluation as well ($p < 0.001$). Dunn's comparisons confirmed that the null model had significantly higher RMSE than all other models for all species, and that the M1, PTT, PTTs, and PA models often outperformed the LIN and DP models across species. No process-based model had outstanding performance relative to other models for this evaluation, as the average RMSE values for all models were within one day of RMSE of one another (Figure A3.3). This being said, the performance of the DP and AT models was more variable over time than other models.

3.2.3. *Predicting and training leaf senescence models with the same moving windows at external validation sites*

When predicting leaf senescence dates at external validation sites over the same 30-year training period, RMSE values were improved relative to the static seven-decade evaluations below. Nonetheless, most models still failed to outperform the null model, demonstrating limited spatial transferability(Fig. 5& Figure A3.5). A Kruskal-Wallis test confirmed differences between models were not significant for both species ($p > 0.05$). Process-based model spatial transferability improved over time for European beech while for horse chestnut it was initially stable and then worsened with time based on RMSE. As with leaf emergence models, spatial transferability over time based on RMSE correlated well with the standard deviation of observations at validation sites over time for European beech and horse chestnut, ($r = 0.93$ - 0.99; $p < 0.001$). The AICc followed a non-linear trend for both species, increasing and then stabilizing or declining slightly. This trend was reflected in the number of observations in each validation window and the residual sum of squares over time underlying the AICc calculation (Figure A3.5).

3.2.4. *Predicting all seven decades of leaf senescence at external validation sites with moving training windows*

When predicting all seven decades of leaf senescence at external validation sites and training on 30-year moving windows, all leaf senescence models predicted leaf senescence dates poorly, demonstrating limited spatial transferability and overall timeseries applicability. A Kruskal-Wallis test confirmed differences between models were significant for both species ($p < 0.05$), though the only significant difference across both species was that the DPDIs outperformed the DM1 model. Unlike for the leaf emergence models, the senescence model performances over time varied by species (Figures A3.6 and A3.7). For European beech, prediction error was stable over time, while for horse chestnut error increased slightly, based on both RMSE and AICc.

3.3. *Calibration trends*

3.3.1. *Leaf emergence models*

While all other parameters were held constant, there were significant trends in most forcing parameter calibration values for leaf emergence models over time (Fig. 6). Within a given species, all forcing parameters tended to follow a similar trend over time, and with a similar trend magnitude, reflecting the influence of species-specific phenology responses rather than general model-level tendencies across species. Forcing parameter trends were greatest in absolute magnitude for horse chestnut and in contrast most stable for European beech, while the trends for other species were intermediate. Among the most pronounced parameter changes based on correlation strength was a 23-day delay in the *t0* accumulation start date parameter with the PA model for horse chestnut (Fig. 6).

In contrast to that of forcing parameters, fewer trends were found in the calibration of optimal chilling parameters, and these trends were more related to the model used rather than the species. However, the directions of trends were consistent within each species. Chilling parameters within the PA and DP models had significant trends for four species, while SQ model parameters did not follow a significant trend

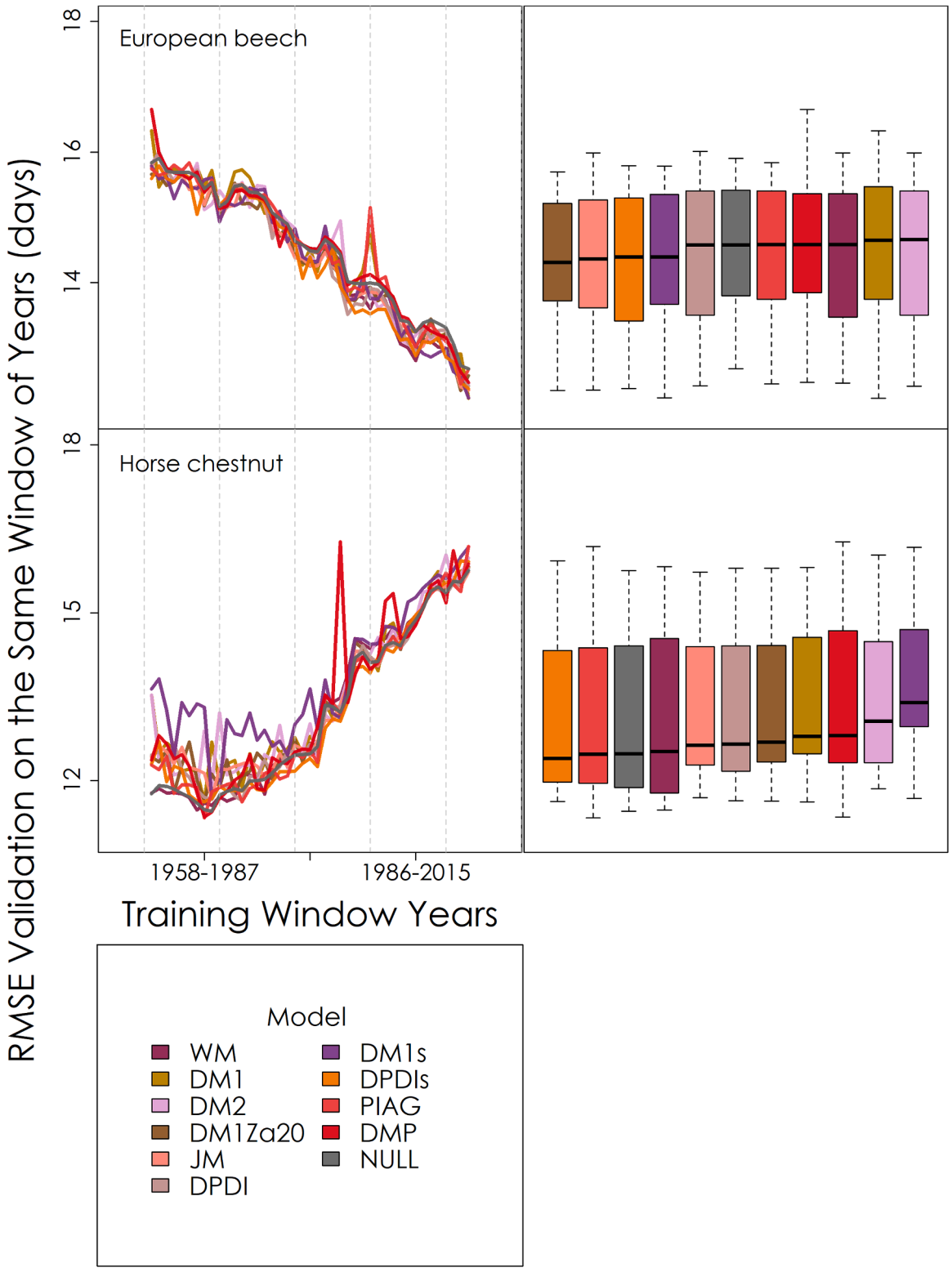


Fig. 5. Line plots showing performance of leaf senescence models with 30-year moving training windows in predicting leaf emergence dates at external validation sites for the same 30-year period, based on root mean squared error. Refer to Table 2 for model abbreviations. Boxplots to the right of each line plot show the overall performance over time for each species-model combination.

over time. Within species, the trends in chilling parameters were consistent with those of forcing parameters, such that both chilling and forcing parameters declined over time for Norway spruce and increased for horse chestnut. For example, earlier start dates and lower critical thresholds for both chilling and forcing accumulation were more optimal for Norway spruce and European larch. The sole independent photoperiod parameter, *f*, for the M1 model, was stable over time for Norway spruce and European beech, two species known to be especially sensitive to photoperiodic constraints (Chiang et al., 2021; Vitasse and Basler, 2013), though varied for other species.

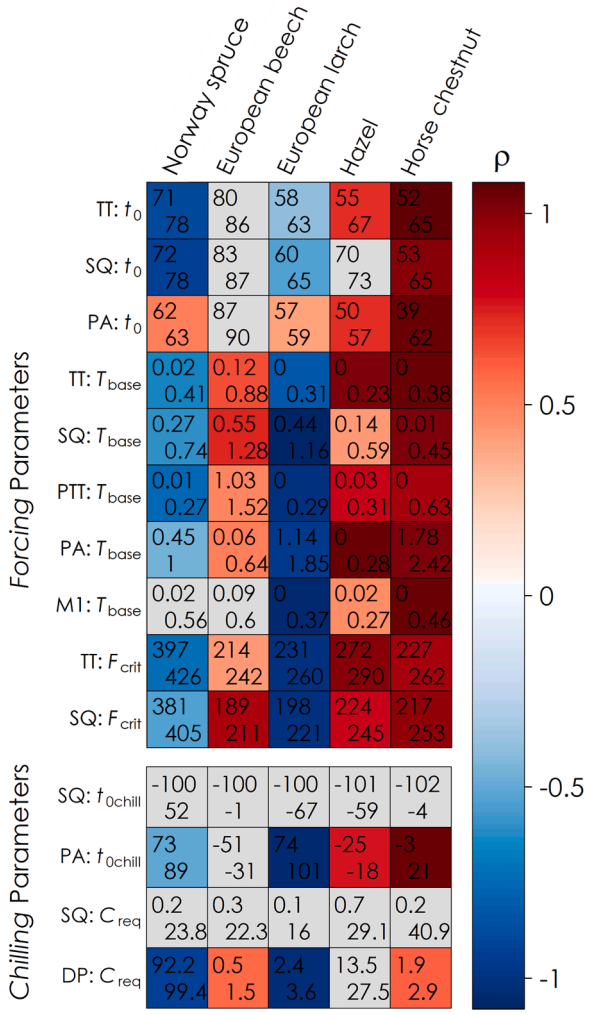


Fig. 6. Spearman correlation (ρ) patterns of calibrated leaf emergence model forcing and chilling parameters in relation to time from 1951–2022 with 30-year moving windows. For each calibration, other model parameters were held constant at optimal values. Model parameters which followed a significant positive trend have a red background, while parameters which followed a significant negative trend have a blue background. Model parameters with non-significant ($p > 0.05$) linear slope terms over time have a grey background. The minimum and maximum optimal values for parameters over time are presented in each cell. Refer to Table 1 for model abbreviations.

3.3.2. *Leaf senescence models*

In contrast to leaf emergence parameters, more than half of all leaf senescence model optimal cooling and photoperiod parameters did not follow significant trends over time (Fig. 7). However, as with leaf emergence forcing parameters, parameter trends were more pronounced within a species rather than at a model level. Horse chestnut parameters often followed significant trends over time, generally increasing in value. In contrast, most parameters were stable over time for European beech, though several decreased over time.

3.4. *Parameter sensitivity*

3.4.1. *Leaf emergence models*

Our sensitivity analyses revealed that the most directly and interactively influential parameters varied across leaf emergence models (Fig. 8), with the onset, rate, and critical threshold for accumulation being most influential in one or more models across the ensemble. The importance of parameter interactions varied across models and

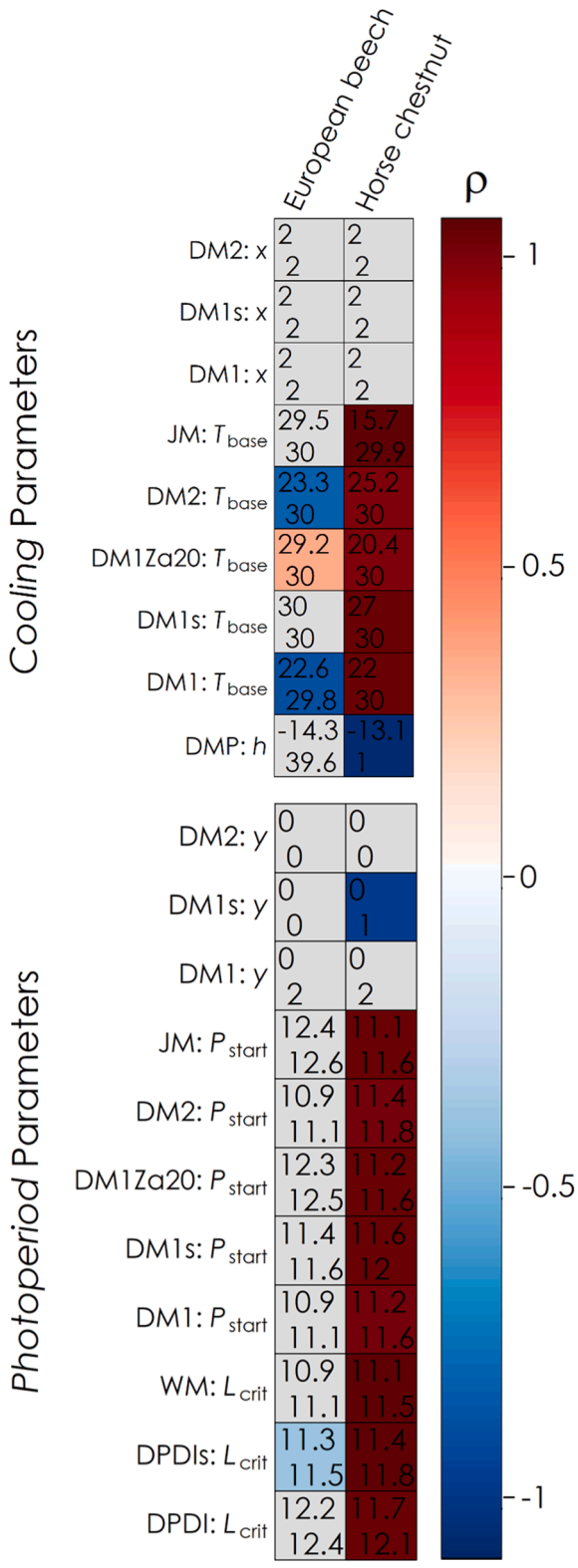


Fig. 7. Spearman correlation (ρ) patterns of calibrated leaf senescence model cooling and photoperiod parameters in relation to time from 1951–2022 with 30-year moving windows. For each calibration, other model parameters were held constant at optimal values. Model parameters which followed a significant positive trend have a red background, while parameters which followed a significant negative trend have a blue background. Model parameters with non-significant ($p > 0.05$) linear slope terms over time have a grey background. The minimum and maximum optimal values for parameters over time are presented in each cell. Refer to Table 2 for model abbreviations.

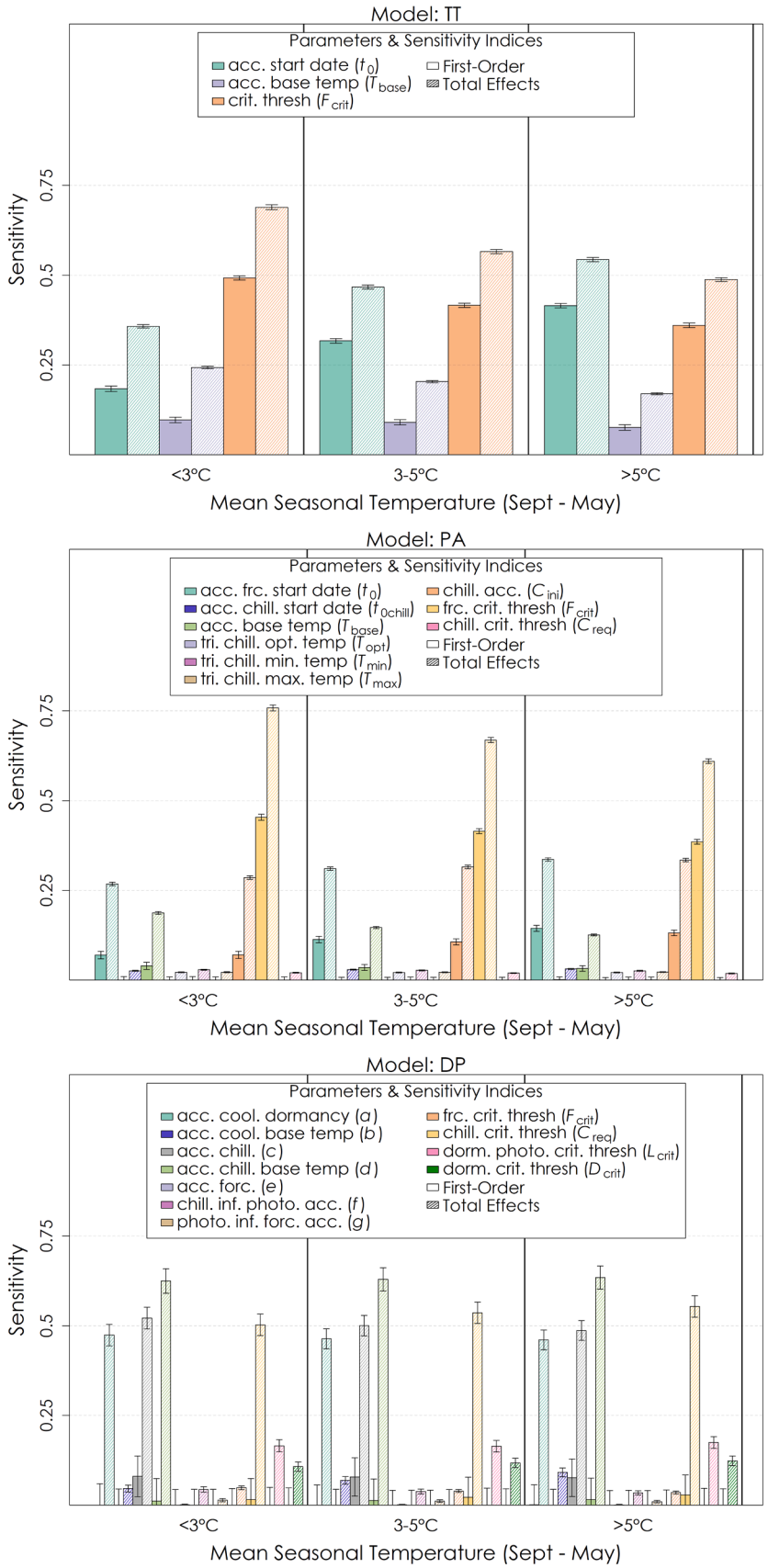


Fig. 8. First-order and total effects sensitivity of one- (ecodormancy only), two- (endodormancy and ecodormancy), and three- (dormancy induction, endodormancy, and ecodormancy) phase leaf emergence model parameters as a function of mean seasonal temperature calculated from September to May. Models shown here include the Thermal Time (TT; top), Parallel (PA; middle), and Dormphot (DP; bottom) model.

generally increased with increasing model complexity. In the simplest ecodormancy models, the interactive influence of model parameters was of minor importance, as the total effects sensitivity was similar to the first-order effect sensitivity across parameters. In more complex models, interactions were more important, with parameters having much greater total effects sensitivity than direct effects.

Direct parameter influences varied between site-years with warmer versus colder winter and spring months (September – May). The TT model showed the most pronounced temperature influence on sensitivity results. For the TT model, F_{crit} was the most directly influential parameter under colder and moderate conditions, while the influence of *t0*, the start date of accumulation, surpassed F_{crit} under warmer conditions (Fig. 8). This suggests that under warm winter-spring seasons, the timing of the onset of ecodormancy may have a greater impact on leaf emergence than the rate or threshold for forcing accumulation. For all other models, parameter sensitivities often varied under different average seasonal temperatures as well, but not enough to change which parameter was the most influential. In the most complex model, DP, parameter sensitivities did not vary notably across different seasonal temperatures.

3.4.2. Leaf senescence models

The most directly influential parameters varied across leaf senescence models, though were often related to critical thresholds for triggering leaf senescence (Fig. 9). Unlike the leaf emergence models, the indirect influence of leaf senescence model parameters had an important influence on all model outcomes, with the total effects often surpassing first-order effects. For three models, the interactive effects of all model parameters were greater than the direct effects of any singular parameters. For all other models, except for JM, two or more parameters exhibited greater interactive effects than direct effects.

The direct effect of all leaf senescence model parameters remained largely stable across different spring and summer period temperatures (May – October), with no one parameter becoming more important under different conditions (Fig. 9). In contrast, total effects sensitivity, including direct and interactive parameter effects, varied under different conditions for two models, the JM and DM1Za20 models, while for all other models, total effects sensitivity was stable across different seasonal temperatures. In the JM, no parameters had important interactive effects under cold and moderate conditions, while under warm conditions the daylength threshold for accumulation, *Pstart*, had important interactive effects. Alternatively, tor the DM1Za20, T_{base} became more influential than *Pstart* under warm conditions.

3.5. Spatial patterns in model performance

3.5.1. Influence of training elevation for leaf emergence models

Model performances varied across different training elevations, with pronounced patterns in model transferability at high and low training elevations (Fig. 10). European beech was the exception, with a 1-day range in RMSE across training elevations. For other species, there was a 7-day to 12-day range in RMSE across training elevations, typically with higher error under training at a higher elevation. More complex models were less transferable from different training elevations, though several simple models also exhibited moderate error at high elevations.

3.5.2. Influence of training elevation for leaf senescence models

For this evaluation, we were interested in whether models were more sensitive to either training or validation at particular elevations. To investigate the influence of training elevation, we systematically separated training data into three elevational bands: low (< 600 m), medium (600–1000 m), and high (>1000 m), and tested models trained with data from these bands on stratified validation data. The performance of leaf senescence models was more consistent across training elevations than it was for leaf emergence models (Fig. 11), with most models performing to a similar degree across all elevations. For several models, such as the DM1s, DM1, and DM2 models, there were high error rates for either high or intermediate elevations. As with the leaf emergence models, leaf senescence models had a narrow 1-day RMSE range in performance across training elevations for European beech, though a much greater RMSE range of 11 days for horse chestnut.

3.5.3. Influence of validation elevation for leaf emergence models

For the investigation of validation elevation, we did the opposite from above – training models with stratified data and then validating with each band: low (< 600 m), medium (600–1,00 m), and high (>1000 m). There was often a lower range in RMSE across validation elevations relative to training elevations, though not for European beech (Fig. 12). For most species, RMSE values ranged by 3–5 days across validation elevations, though for horse chestnut there was a 10-day RMSE range. No validation elevation corresponded with consistently high error across species and models, with high validation elevations being best for European beech and horse chestnut, while low or intermediate validations were best for others. More complex models were generally less robust to different validation elevations compared to simpler models. The differences between models were more evident with different training elevations than validation elevations, indicating that these models are more robust to different validation elevations rather than training elevations.

3.5.4. Influence of validation elevation for leaf senescence models

In contrast to the training elevation investigation, the performance of leaf senescence models varied more as a function of validation elevation than model (Fig. 13). For both species, most models performed better with validations at high elevations. There were divergent patterns between species in the relative range in RMSE across validation elevations compared to training elevations. For European beech and horse chestnut, there was a 2–3 day range in RMSE across validation elevations, which was similar or greater than the range across training elevations. For both species, validation with low elevations generally led to higher RMSE values across all models, and European beech also had higher RMSE values for intermediate elevations. Several models were distinct from others at a given validation elevation, with higher error for the PIAG and several other models at higher elevations.

4. Discussion

4.1. Key findings

Here, we conducted a novel assessment of the stability and biological relevance of a wide range of process-based models using an extensive leaf phenology dataset spanning over seven decades and 1700 m elevation in Switzerland. Our findings suggest that plant responses to processes driving leaf emergence have converged over time within species, enhancing the transferability of leaf emergence models – contrary to our expectations, though in line with a previous study finding warming leads to reduced variability in phenology (Denéchère et al., 2021). Conversely, leaf senescence models frequently struggled, or narrowly succeeded, in outperforming the null model predicting the mean senescence date of the training dataset. For leaf senescence, we found mixed trends in plant responses between species, corresponding to degraded transferability of senescence models for horse chestnut and stable or improved transferability for European beech. The temporal patterns in optimal values we observed suggest that driver influences are shifting uniquely for each species. Cold-climate suited species are exhibiting earlier and lower thresholds for the accumulation of spring warming effects, while warmer-climate suited species show the opposite trend.

Our sensitivity analyses suggest that with warming, chilling may become increasingly important in driving leaf emergence, as parameters related to the completion of endodormancy and onset of ecodormancy became increasingly important under warmer conditions for several

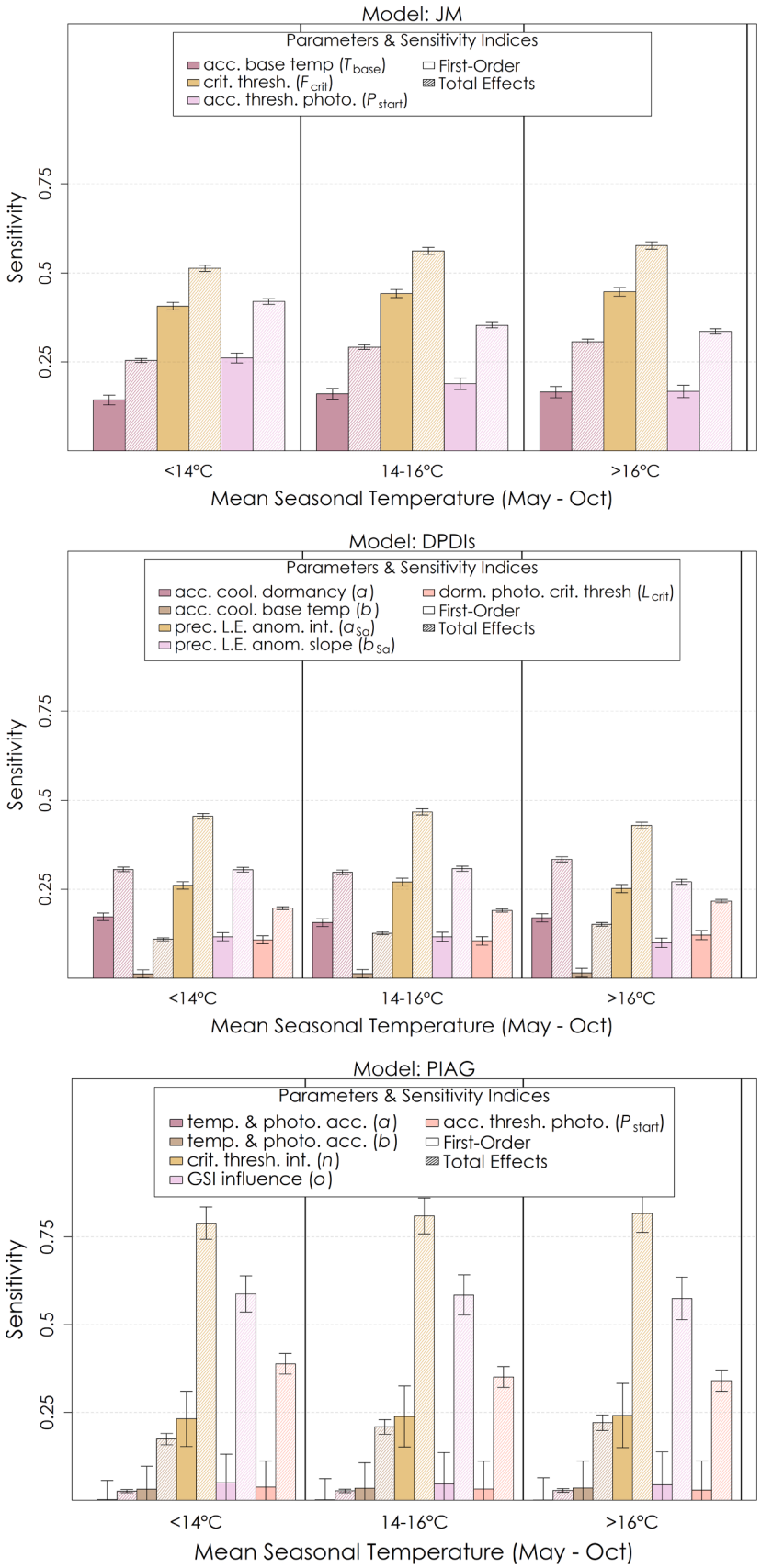


Fig. 9. First-order and total effects sensitivity of leaf senescence model parameters as a function of mean seasonal temperature calculated from May to October. Models shown here include the Jeong (JM; top), Dormphot Dormancy Induction with Preceding Spring Influence (DPDIs; middle), and Photosynthesis-Influenced Autumn with Growing Season Index (PIAG; bottom) model.

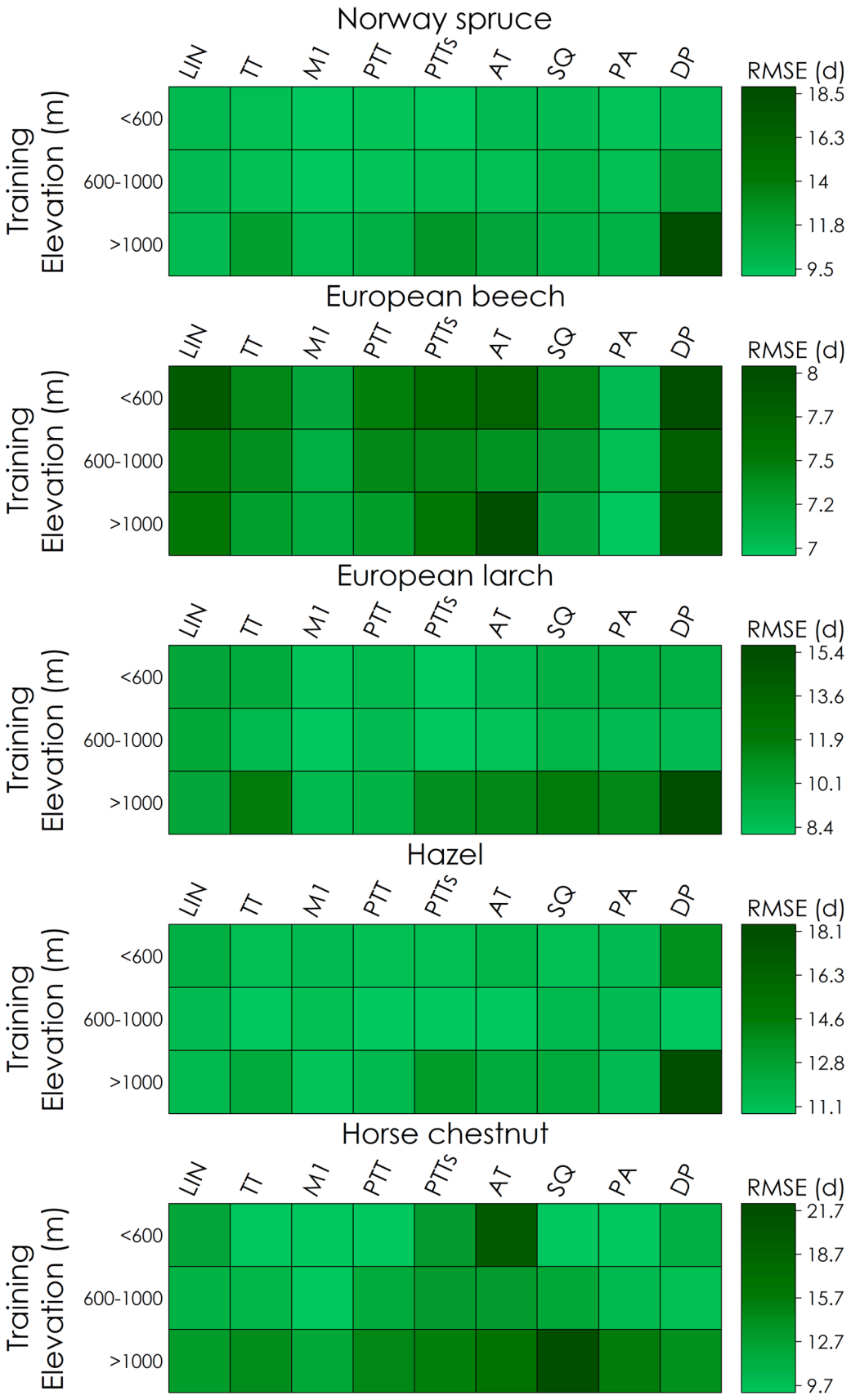


Fig. 10. Performance of leaf emergence models in predicting leaf emergence dates for all species at stratified external validation sites, as a function of training elevation. Refer to Table 1 for model abbreviations.

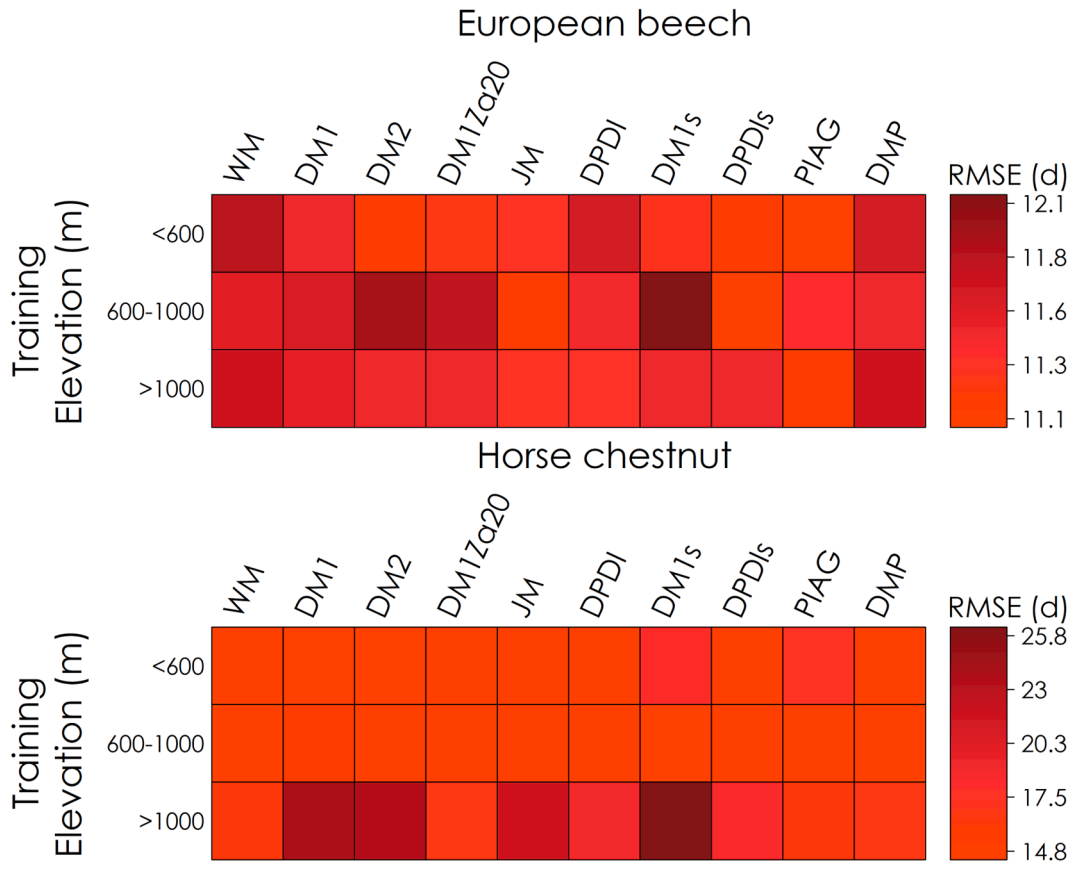


Fig. 11. Performance of leaf senescence models in predicting leaf senescence dates for all species at stratified external validation sites, as a function of training elevation. Refer to Table 2 for model abbreviations.

models. In contrast, our sensitivity analyses for leaf senescence models suggest that either the relative importance of senescence drivers will remain stable with warming, or current models fail to capture the influence of processes driving long-term patterns in leaf senescence. Our spatial investigations indicated that most complex models were more sensitive to training at different elevations rather than validation at different elevations, suggesting that phenology response processes may vary non-linearly with elevation due to local adaptations, and complex models trained at one elevation may be less transferable to others. Future work should prioritise refining leaf senescence models, with special attention to incorporating novel driving mechanisms, and improving transferability over time.

4.2. Trends in plant phenology responses over time

Contrary to expectations given the rapid pace of ongoing phenological change, leaf emergence model performance remained stable over time and across sites, showing no decline in recent decades and consistently outperforming the null model. Models for some species even demonstrated improved spatial transferability over time. However, this was well correlated with the declining variability in leaf emergence at validation sites and improved performance of the null model over time (Figure A3.2). This homogenization is consistent with Denéchère et al. (2021), who reported reduced interpopulation budburst timing variability during warmer springs. In contrast, leaf senescence models showed poor performance and limited transferability, with trends varying by species. Spatiotemporal transferability declined over time for horse chestnut and improved for European beech. This declining spatial transferability also correlated with increasing variability in leaf senescence at validation sites for horse chestnut over time, potentially related to the horse-chestnut leaf miner, and conversely with declining variability for European beech. These divergent trends in the transferability of leaf emergence and senescence models over time indicate varied impacts of climate change since the mid-20th century on leaf emergence and senescence response processes. Leaf emergence timings are becoming increasingly consistent in response to drivers, with warmer springs becoming increasingly common, while trends in leaf senescence responses to drivers show a mixed pattern between species. With continued climate change and increasing climate variability, it is also likely that spring leaf emergence may exhibit pronounced variability for unusually warm or cold winters and springs (Denéchère et al., 2021; Liu and Zhang, 2020).

4.3. Improving the biological plausibility of models

While all leaf emergence models outperformed the null model, their very similar performance despite strongly differing mechanistic representations leads us to question the biological plausibility of simulated model processes. For example, across our 1.5 to 14.1 °C annual average temperature gradient, models which include chilling and/or the influence of photoperiod on leaf emergence should diverge from simpler thermal models such as the TT model. More replicated experiments across driver space are needed to refine the response of leaf emergence and sub-processes to chilling, photoperiod, and forcing, and to better quantify species-specific mechanisms and validate the biological realism of model parameters (Gao et al., 2024; Hänninen et al., 2019; Wolkovich et al., 2022; Zhang et al., 2022). Additionally, these experiments would help to better constrain reasonable parameter ranges for each species and model, improving the efficiency of future process-based modelling efforts. Further, exploring novel drivers and mechanisms may enhance the biological realism of future models.

All leaf senescence models performed poorly, often failing to

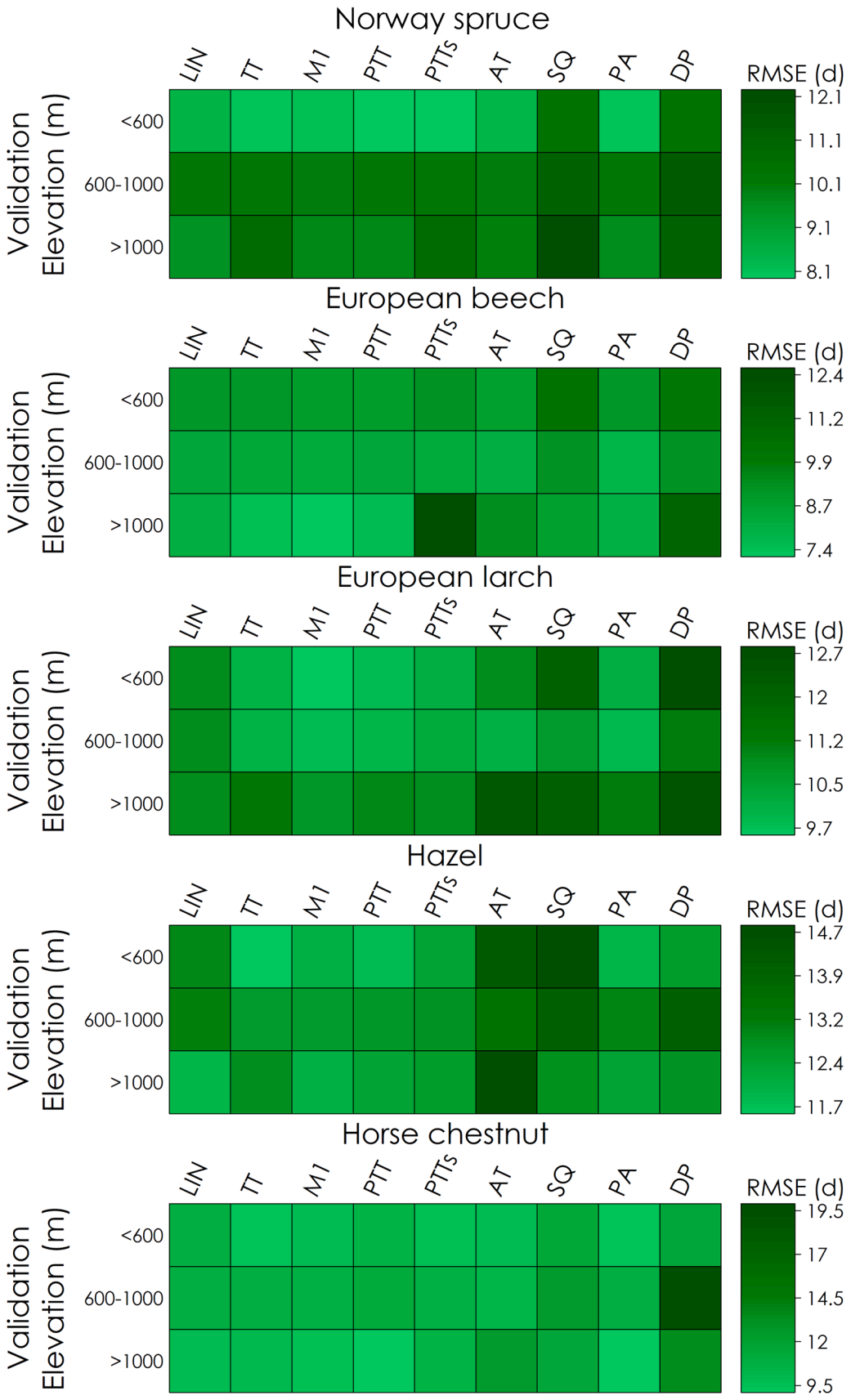


Fig. 12. Performance of leaf emergence models trained across stratified sites in predicting leaf emergence dates for all species, as a function of validation elevation. Refer to Table 1 for model abbreviations.

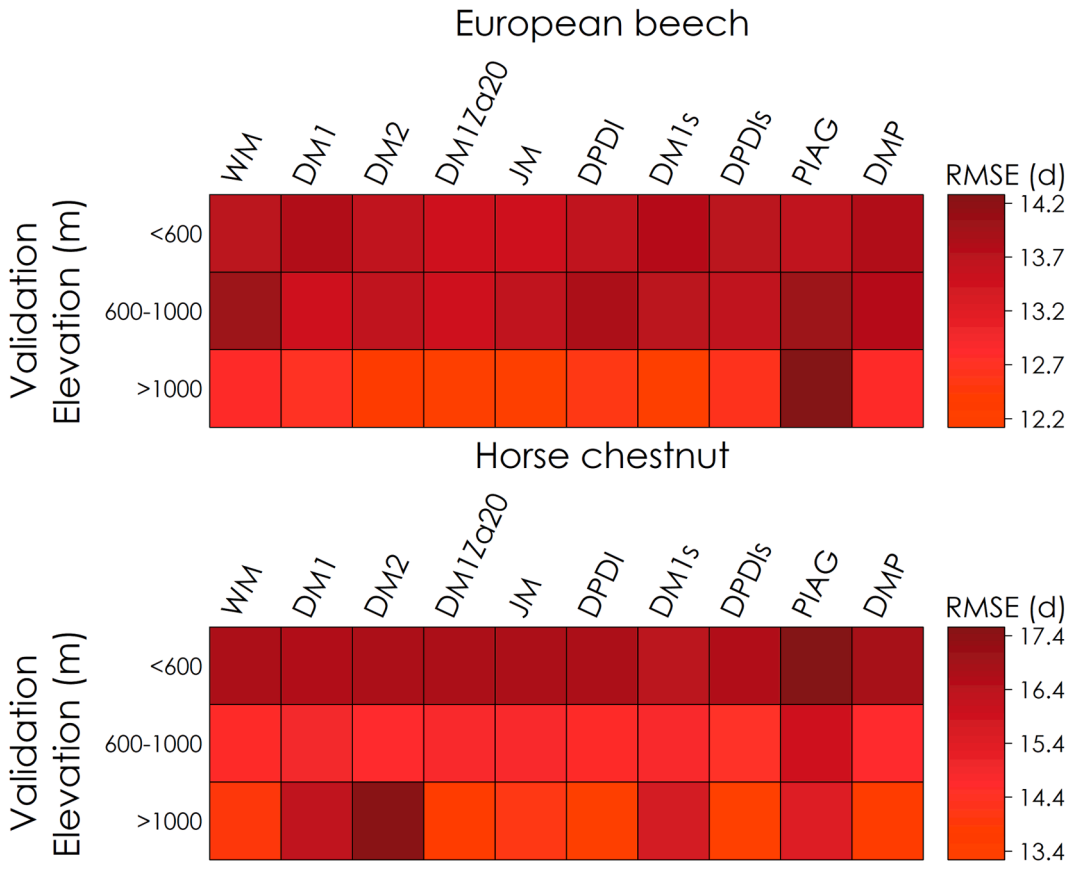


Fig. 13. Performance of leaf senescence models trained across stratified sites in predicting leaf senescence dates for all species, as a function of validation elevation. Refer to Table 2 for model abbreviations.

outperform the null model which assumes a constant date based on the mean of training data, demonstrating limited predictive skill. While few studies compare the performance of autumn process-based models to the null model, the two to three weeks RMSE we observed is consistent with other studies (Delpierre et al., 2009; Liu et al., 2020; Meier and Bigler, 2023). We found that senescence models performed better with validation at high elevations, where climatic variables such as cooling temperatures play a large role, while at other elevations where drought, internal development, and nutrient levels may play a larger role, predictions were more challenging (see also Bigler and Vitasse, 2021; Estiarte and Peñuelas, 2015). Given the poor performance of senescence models, we investigated whether the performance of the autumn phenology models could be improved with more replicated calibrations (60,000 – 100,000 iteration allowances). We found this more replicated calibration led to worsening predictions at external validation sites relative to the 40,000 iterations typically used. Future process-based modelling studies and software development would benefit from considering the number of calibration iterations itself as a hyperparameter for each model calibration, a technique commonly used in machine learning, to avoid both over-fitting and incomplete optimizations for each model (Bischl et al., 2023; Bishop and Nasrabadi, 2006; Hastie et al., 2009).

Including novel drivers like precipitation and estimated productivity and the timing of preceding leaf emergence did not improve predictions, unlike some previous studies (Fu et al., 2014; Keenan and Richardson, 2015; Liu et al., 2020; Zani et al., 2020; Zohner et al., 2023), though in line with Meier and Bigler (2023). A few promising avenues to improvement may be to explore different mechanisms of incorporating these senescence drivers or entirely different drivers. For example, soil moisture, nutrient levels, and light intensity are known to influence leaf senescence (Delpierre et al., 2017; Estiarte and Peñuelas, 2015; Vitasse et al., 2021), though these drivers are not commonly included. Additionally, Bigler and Vitasse (2021) show that episodic drought and heat spells can cause premature leaf discolouration, which will likely become increasingly common with continued global warming. This could compound current challenges with process simulations of leaf senescence, as these events are not affected by the same drivers as general historical leaf senescence patterns. Consequently, future model exploration may benefit from including mechanisms for both episodic and legacy effects, as well as exploring novel drivers.

The poor performance of senescence models relative to emergence models may also stem in part from observational challenges, as leaf colour change occurs more gradually than emergence (Liu et al., 2021; Richardson et al., 2018a; Sonnentag et al., 2012). A recent modelling study also found process-based models were often outperformed by a null model and reported that high model errors were likely due to noise in visual observations (Meier et al., 2025). Previously, an investigation of inter-observer variability revealed that observer uncertainty is nearly doubled for leaf senescence relative to budburst (Liu et al., 2021). Automated observational techniques, such as stationary timelapse photography in the form of phenocams, may help to capture more precise quantifications of gradual subtle changes in leaf colouration (Seyednasrollah et al., 2019). Additionally, the narrow latitudinal range for our study limited variation in the key autumn phenology driver of photoperiod (Caffarra et al., 2011), and a study with some of the same species and models with a broader latitudinal range reported slightly improved performance by about 2 days RMSE (Liu et al., 2020). Further, the lower null model RMSE for leaf senescence models relative to emergence models poses a greater challenge to outperform, and the mean date in the null model implies a photoperiodic constraint of sorts. Despite the quality control steps outlined in Auchmann et al. (2018), the high error of models for horse chestnut may have been influenced by the

outbreak of the horse-chestnut leaf miner *Cameraria ohridella* in Switzerland since 1999 which can cause premature leaf colouration in highly infested stands (Kopacka and Zemek, 2017; Krivosheina and Ozerova, 2020).

4.4. Trends in optimal parameter values

The optimal parameter values for leaf emergence and senescence models varied significantly over time for most species except for European beech. This instability may partially stem from the calibration failing to fully capture driver influences over the seven-decade training period, leading to unstable parameters. However, the stable parameters found for photoperiod-sensitive European beech (Vitasse and Basler, 2013) and divergent parameter trends between species suggest that driver influences are shifting uniquely for each species. For cold-adapted species like European larch and Norway spruce, earlier onsets and lower thresholds for forcing and chilling accumulation were more suitable over time. This matches our expectations, as these species are common at rather high elevations, where chilling requirements are readily satisfied, allowing earlier forcing accumulation, and consequently leaf emergence is advancing more rapidly than at lower elevations (Vitasse et al., 2018b). These earlier onsets and lower thresholds for forcing and chilling accumulation may increase susceptibility to late spring frost damage of newly formed leaf tissues, which is also consistent with experimental and observational studies (Richardson et al., 2018b; Vitasse et al., 2018a). Conversely, the later and higher thresholds for forcing and chilling accumulation we observed for warm-climate suited species like horse chestnut and hazel may correspond to reduced frost exposure, though with potential trade-offs in heightened resource competition (Baumgarten et al., 2023). Therefore, these trends may reduce the fitness of cold-adapted species and increase the fitness of warm-adapted species by altering frost damage risk. At the same time, these trends confer advantages to cold-adapted species over warm-adapted species by allowing for earlier emergence and reduced resource competition. The distinct species-specific parameter trends we found aligns with a recent study which found that dormancy release responses can vary more between different species at the same site than between individuals of the same species at different sites (Walde et al., 2024). Consequently, future modelling efforts should consider these unique species-level responses to phenology drivers.

For leaf senescence model parameters, European beech showed stable parameters over time, while increasing base temperature and photoperiod accumulation thresholds were optimal for horse chestnut. This apparent increase in critical photoperiod parameters is generally not expected unless there is acclimation related to warming conditions. Overall, the model parameter space included herein for leaf senescence models failed to accurately capture and explain the advancement in leaf senescence over time for horse chestnut, which may be a further indication of the influence of non-climatic effects on leaf senescence for horse chestnut.

4.5. Influential parameters

Our sensitivity analyses revealed that the most influential parameters for leaf emergence models varied under different conditions and model types, with the critical threshold for leaf emergence being most important for some models, and the onset or rate of forcing accumulation being most important for others. Complex models were more influenced by chilling-related parameters and parameter interactions, consistent with Gauzere et al. (2019). Changes in emergence parameter sensitivities under different conditions suggest that models trained in colder conditions may be less applicable to warmer conditions due to non-linear influences of drivers across a temperature gradient, challenging future predictive modelling in the context of climate change. Despite this, our temporal evaluations above demonstrated stable or improved transferability in leaf emergence models over seven decades, though complex models including more detailed endodormancy representations were indeed less transferable. The increasing importance of endodormancy completion and ecodormancy onset parameters with warming also aligns with Gauzere et al. (2019), as during warmer winters delays in endodormancy completion may constrain leaf emergence. This is supported by the greater emergence advancement in response to warming observed at high elevations with ample chilling exposure relative to low elevations by Vitasse et al. (2018b).

Parameters with optimal values which varied over time were also found in many cases to be most influential on leaf emergence model outcomes, contrary to our expectation that influential parameters would not exhibit significant calibration trends. For example, the critical threshold for leaf emergence, F_{crit}, captures the cumulative effects of drivers in several models and followed a significant trend over time for all species and models. This threshold parameter increased for warm-climate species, potentially due to constraints like photoperiod limiting further advances in leaf emergence, while it decreased for cold-climate species, likely the result of greater emergence advancement in response to warming being possible under high-elevation conditions with more ample chilling. Parameters which had greater interactive effects also varied over time. These significant trends in optimal parameter values are therefore important to evaluate, as in many cases these trending parameters may be highly influential on model outcomes. Further, these trends in sensitivity and optimal values indicate phenology response processes are shifting over time, with important implications for long-term projections.

For leaf senescence models, parameter influences were more stable across growing conditions, and interactions were important for almost all models. Given the poor performance of leaf senescence models herein, the sensitivity analysis results require further consideration. Parameter sensitivities were stable across warm versus cold conditions, and parameters related to photoperiod were often less influential than other parameters, in contrast to expectations based on previous studies (Chen et al., 2020; Fu et al., 2018; Gill et al., 2015; Way and Montgomery, 2015).

5. Conclusion

With a dataset that spanned an extensive time and elevational gradient, we examined the stability and biological relevance of widely used plant phenology models. Leaf emergence models generally showed improved transferability and robustness over time, with emergence dates converging across sites. However, the overall similar performance of different emergence models across species and conditions raises questions about the reliability of these models, particularly in their ability to simulate chilling and photoperiod influences. Senescence models instead exhibited divergent trends, often failing to outperform the null model, and demonstrate a strong need for further refinement. Some promising avenues for improving leaf senescence models may be incorporating additional drivers like soil moisture, nutrient levels, and solar radiation and exploring mechanisms for episodic and legacy effects. Trends in optimal parameter values for emergence models varied over time, with earlier and lower thresholds for the accumulation of warming effects over time for species typical at high elevations, likely due to ample chilling exposure for these species. Warm-climate species instead required higher thresholds, potentially due to constraining influences like photoperiod. Sensitivity analyses revealed increasing influence of endodormancy completion and ecodormancy onset parameters with warming, demonstrating the non-linear influence of emergence drivers across temperature gradients. Notably, several highly influential parameters also showed significant trends in optimal values over time. Future modelling studies should account for species-specific trends as well as model-specific trends in parameter sensitivity with warming. Our study demonstrates the continued value of long-term citizen science efforts and process-based modelling in phenology research, which together are crucial to understanding plant phenology

responses to climate change.

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CRediT authorship contribution statement

Lynsay Spafford: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **J. Jelle Lever:** Writing – review & editing, Validation, Project administration, Methodology, Funding acquisition, Conceptualization. **Arthur Gessler:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Roman Zweifel:** Writing – review & editing, Project administration, Funding acquisition. **Barbara Pietragalla:** Writing – review & editing, Data curation. **Jan Dirk Wegner:** Writing – review & editing, Project administration, Funding acquisition. **Vivien Sainte Fare Garnot:** Writing – review & editing. **Christian Sigg:** Writing – review & editing, Project administration, Funding acquisition. **Yann Vitasse:** Writing – review & editing, Validation, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.agrformet.2025.110685.

Data availability

I have shared a link to my data and code at the Attach File step.

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